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Temperature controlled short duration ablation with multiple electrodes

Abstract

Apparatus, including a catheter configured to be inserted into an organ of a human body. A plurality of electrodes are deployed on the catheter, the electrodes being configured to transfer radiofrequency (RF) ablation energy to tissue of the organ. The apparatus also includes a power supply configured to supply the RF ablation energy at a level of up to 100 W to each of the plurality of electrodes simultaneously, so as to ablate respective sections of the tissue of the organ in contact with the electrodes.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a divisional application of U.S. application Ser. No. 15/994,459, filed May 31, 2018, now U.S. Pat. No. 11,666,379, which claims the benefit of U.S. Provisional Patent Application 62/529,158, filed Jul. 6, 2017, both of which are incorporated herein by reference.

FIELD OF THE INVENTION

(1) This invention relates generally to surgery, and specifically to surgery using radiofrequency ablation.

BACKGROUND OF THE INVENTION

(2) Radiofrequency (RF) ablation is a treatment modality that kills unwanted tissue by heat. Starting with cardiac arrhythmia treatment in the 1980s, RF ablation has found clinical application in a number of diseases, and is now the treatment of choice for certain types of cardiac arrhythmia, and certain cancers. During RF ablation, an electrode is inserted into proximity with the target region under medical imaging guidance. Tissue surrounding the electrode in the target region is destroyed by heating via RF electric current.

(3) U.S. Patent Application 2003/0236455 to Swanson et al., describes a probe assembly for mapping and ablating pulmonary vein tissue. The probe assembly includes an expandable and collapsible basket assembly having multiple splines. One or more of the splines carry one or more electrodes adapted to sense electrical activity in the pulmonary vein tissue.

(4) U.S. Patent Application 2014/0066921 to Coe et al., describes balloon catheter neuromodulation systems. The application refers to modulating (e.g., disrupting, ablating,

stimulating) the nerves by mechanical compression, energy delivery, or fluid delivery.

(5) U.S. Pat. No. 5,931,835 to Mackey, describes a radio frequency energy delivery system for multipolar electrode catheters. It is stated that the power, voltage, or temperature delivered to multiple electrodes may be dynamically controlled.

(6) EP Patent Application 1,645,234 to Buysse et al., describes an electrosurgical system employing multiple electrodes. The system employs multiple electrodes for producing large ablation volumes in tissue.

(7) U.S. Patent Application 2002/0161361 to Sherman et al., describes an RF ablation system using electrodes and having automatic temperature control. It is stated that a select number of the electrodes have a temperature sensing device associated with them for providing a temperature signal indicative of the temperature at the interface between the electrode and tissue.

(8) U.S. Patent Application 2001/0020166 to Daly et al., describes a system for simultaneous unipolar multi-electrode ablation. The system is stated to ablate tissue using unipolar RF energy simultaneously delivered to multiple electrodes.

(9) U.S. Pat. No. 6,319,249 to Töllner, describes an ablation catheter with, inter alia, a plurality of ablation electrodes, at least one energy source, and switching elements for connecting the electrodes to the energy source.

(10) U.S. Patent Application 2008/0161797 to Wang et al., describes ablation catheter electrodes having multiple thermal sensors. The electrodes are stated to contain two or more thermal sensors at different positions within the electrode.

(11) Documents incorporated by reference in the present patent application are to be considered an integral part of the application except that, to the extent that any terms are defined in these incorporated documents in a manner that conflicts with definitions made explicitly or implicitly in the present specification, only the definitions in the present specification should be considered.

SUMMARY OF THE DISCLOSURE

(12) An embodiment of the present invention provides apparatus, including: a catheter configured to be inserted into an organ of a human body; a plurality of electrodes deployed on the catheter, the electrodes being configured to transfer radiofrequency (RF) ablation energy to tissue of the organ; and a power supply configured to supply the RF ablation energy at a level of up to 100 W to each of the plurality of electrodes simultaneously, so as to ablate respective sections of the tissue of the organ in contact with the electrodes.

(13) In a disclosed embodiment the plurality of electrodes includes up to twelve electrodes, and the power supply is configured to provide up to 1.2 kW of radiofrequency power.

(14) In a further disclosed embodiment the apparatus includes a plurality of temperature sensors each coupled to measure a respective temperature of one of the plurality of electrodes, and the power supply is configured, when the respective temperature of the one of the plurality of electrodes in contact with one of the sections of tissue exceeds a selected maximum temperature, to reduce the level of power of the RF ablation energy supplied to the one of the plurality of electrodes.

(15) In a yet further disclosed embodiment the apparatus includes a processor which is coupled to the power supply and which is configured to simultaneously measure a respective impedance to the RF ablation energy for each of the plurality of electrodes, and, when a change in the impedance to one of the plurality of electrodes in contact with one of the sections of tissue exceeds a preset value, to halt supply of the RF ablation energy from the power supply to the one of the plurality of electrodes.

(16) The catheter may include a balloon catheter. Alternatively, the catheter may include a basket catheter.

(17) There is further provided, according to an embodiment of the present invention, apparatus, including: a catheter configured to be inserted into an organ of a human body; a first electrode and a second electrode deployed on the catheter, the electrodes being configured to transfer

radiofrequency (RF) ablation energy to tissue of the organ; a first temperature sensor coupled to measure a first temperature of the first electrode; a second temperature sensor coupled to measure a second temperature of the second electrode; a power supply configured to provide the RF ablation energy; a switch connected to the power supply and configured to direct the RF ablation energy to one of the first and second electrodes; and a processor configured, while the power supply is providing the RF ablation energy via the switch to the first electrode, to monitor the first and second temperatures and, responsively to the monitored temperatures, to toggle the switch so as to direct the RF ablation energy to one of the first and second electrodes.

(18) In an alternative embodiment the processor is configured, upon sensing that the first temperature exceeds a predefined ablation temperature threshold while the second temperature does not exceed the ablation temperature threshold, to toggle the switch so as to direct the RF ablation energy to the second electrode.

(19) In a further alternative embodiment the processor is configured to monitor a first time for ablation via the first electrode and a second time for ablation via the second electrode, and, responsively to the monitored times, to toggle the switch so as to direct the RF ablation energy to one of the first and second electrodes. Typically, the processor is configured, upon sensing that the first time for ablation equals or exceeds a preset first time for ablation for the first electrode while the second time for ablation is less than a preset second time for ablation for the second electrode, to toggle the switch so as to direct the RF ablation energy to the second electrode.

(20) In a yet further alternative embodiment the first electrode is configured to transfer the RF ablation energy at a first power level, and the second electrode is configured to transfer the RF ablation energy at a second power level, and the power supply is configured to supply power to the electrodes at a level no greater than a maximum of the first and second power levels.

(21) The catheter may be a balloon catheter. Alternatively, the catheter may be a basket catheter.

(22) There is further provided, according to an embodiment of the present invention, a method, including: inserting a catheter into an organ of a human body; deploying a plurality of electrodes on the catheter, the electrodes being configured to transfer radiofrequency (RF) ablation energy to tissue of the organ; and supplying with a power supply the RF ablation energy at a level of up to 100 W to each of the plurality of electrodes simultaneously, so as to ablate respective sections of the tissue of the organ in contact with the electrodes.

(23) There is further provided, according to an embodiment of the present invention, a method, including: inserting a catheter into an organ of a human body; deploying a first electrode and a second electrode on the catheter, the electrodes being configured to transfer radiofrequency (RF) ablation energy to tissue of the organ; coupling a first temperature sensor to measure a first temperature of the first electrode; coupling a second temperature sensor to measure a second temperature of the second electrode; configuring a power supply to provide the RF ablation energy; connecting a switch to the power supply and configuring the switch to direct the RF ablation energy to one of the first and second electrodes; and while the power supply is providing the RF ablation energy via the switch to the first electrode, monitoring the first and second temperatures and, responsively to the monitored temperatures, toggling the switch so as to direct the RF ablation energy to one of the first and second electrodes.

(24) The present disclosure will be more fully understood from the following detailed description of the embodiments thereof, taken together with the drawings, in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic illustration of an invasive medical procedure using apparatus, according to an embodiment of the present invention;

- (2) FIG. 2 is a schematic perspective view of a balloon catheter in its inflated configuration, according to an embodiment of the present invention;
- (3) FIG. 3 is a schematic view of the balloon catheter deployed in a pulmonary vein, according to an embodiment of the present invention;
- (4) FIG. 4 is a schematic view of a plurality of leaves of flexible circuit assemblies, according to an embodiment of the present invention;
- (5) FIG. 5 is a schematic perspective view of a flexible circuit assembly partly lifted from the balloon catheter according to an embodiment of the present invention;
- (6) FIG. 6 is a block diagram of the apparatus of FIG. 1, according to a first embodiment of the present invention;
- (7) FIG. 7 is a flowchart of steps of an algorithm performed in operation of the first embodiment;
- (8) FIG. 8 is a block diagram of the apparatus of FIG. 1, according to a second embodiment of the present invention; and
- (9) FIG. 9 is a flowchart of steps of an algorithm performed in operation of the second embodiment.

DETAILED DESCRIPTION OF EMBODIMENTS

(10) Overview

(11) In embodiments of the present invention, a catheter, having a plurality of electrodes deployed thereon, is inserted into an organ, typically the heart, of a human body. The electrodes are configured to transfer radiofrequency (RF) ablation energy to tissue of the organ.

(12) In a first embodiment of the invention, a power supply supplies the RF ablation energy at a level of up to 100 W to each of the plurality of electrodes simultaneously, so as to ablate respective sections of the tissue of the organ in contact with the electrodes. During such simultaneous ablation, careful monitoring of the temperature and impedance at each electrode separately allows embodiments of the present invention to perform multiple tissue ablations at the powers of up to 100 W at each electrode without adverse effects on the tissue. The high powers enable an overall ablation session for the multiple ablations to be shortened to times typically of no more than 10 s.

(13) In a second embodiment of the invention, the plurality of electrodes comprise a first electrode and a second electrode. A first temperature sensor measures a first temperature of the first electrode, and a second temperature sensor measures a second temperature of the second electrode. A power supply provides the RF ablation energy, and a switch is connected to the power supply and is configured to direct the RF ablation energy to one of the first and second electrodes. A processor is configured, while the power supply is providing the RF ablation energy via the switch to the first electrode, to monitor the first and second temperatures and, responsively to the monitored temperatures, to toggle the switch so as direct the RF ablation energy to one of the first and second electrodes. Switching the RF energy between the electrodes, depending on the temperatures of the electrodes, ensures efficient utilization of a power supply that may be unable to provide high powers to both electrodes simultaneously, due to a maximum power rating of the power supply.

DETAILED DESCRIPTION

(14) In the following description, like elements in the drawings are identified by like numerals, and like elements are differentiated as necessary by appending a letter to the identifying numeral.

(15) FIG. 1 is a schematic illustration of an invasive medical procedure using apparatus 12, according to an embodiment of the present invention. The procedure is performed by a medical professional 14, and, by way of example, the procedure in the description hereinbelow is assumed to comprise ablation of a portion of a myocardium 16 of the heart of a human patient 18. However, it is understood that embodiments of the present invention are not merely applicable to this specific procedure, and may include substantially any procedure on biological tissue or on non-biological materials.

(16) In order to perform the ablation, medical professional 14 inserts a probe 20 into a sheath 21 that has been pre-positioned in a lumen of the patient. Sheath 21 is positioned so that a distal end

22 of probe 20 enters the heart of the patient. A balloon catheter 24, which is described in more detail below with reference to FIGS. 2-5, is deployed through a lumen 23 of the probe 20, and exits from distal end 22 of the probe 20.

(17) As shown in FIG. 1, apparatus 12 is controlled by a system processor 46, which is located in an operating console 15 of the apparatus. Console 15 comprises controls 49 which are used by professional 14 to communicate with the processor. During the procedure, the processor 46 typically tracks a location and an orientation of the distal end 22 of the probe 20, using any method known in the art. For example, processor 46 may use a magnetic tracking method, wherein magnetic transmitters 25X, 25Y and 25Z external to patient 18 generate signals in one or more coils positioned in the distal end of the probe 20. The CARTO® system available from Biosense Webster, of 33 Technology Drive, Irvine, CA 92618, uses such a tracking method.

(18) The software for the processor 46 may be downloaded to the processor in electronic form, over a network, for example. Alternatively or additionally, the software may be provided on non-transitory tangible media, such as optical, magnetic, or electronic storage media. The tracking of the distal end 22 is typically displayed on a three-dimensional representation 60 of the heart of patient 18 on a screen 62.

(19) In the description herein processor 46 is assumed to be formed from any suitable integrated circuits, including, but not limited to, an ASIC (application specific integrated circuit), an FPGA (field-programmable gate array), an MCU (microcontroller unit), and a CPU.

(20) In order to operate apparatus 12, the processor 46 communicates with a module bank 50, which has a number of modules used by the processor to operate the apparatus. Thus, the bank 50 comprises a temperature module 52, a power supply 54, a switch 57, an irrigation module 55, and an electrocardiograph (ECG) module 56, the functions of which are described below. Bank 50 typically comprises other modules, such as a force module for measuring the force on the distal end 22, and a tracking module for operating the tracking method used by the processor 46. For simplicity, such other modules are not illustrated in FIG. 1. The modules may comprise hardware as well as software elements.

(21) FIG. 2 is a schematic perspective view of the balloon catheter 24 in its inflated configuration, and FIG. 3 is a schematic view of the balloon catheter deployed in a pulmonary vein, according to an embodiment of the present invention. In a disclosed embodiment, where the balloon catheter 24 is used to ablate an ostium 11 of a lumen, such as a pulmonary vein 13, as shown in FIG. 3, the balloon catheter 24 is supported by a tubular shaft 70 having a proximal shaft portion 82 and a distal shaft end 88. The shaft 70 comprises a hollow central tube 74, which permits a support catheter to pass therethrough and through the distal shaft end 88. The support catheter may be a focal linear catheter or a lasso catheter 72, as illustrated. The lasso catheter 72 may be inserted into the pulmonary vein (PV) to position the balloon catheter 24 correctly with respect to the ostium prior to ablation of the ostium. The distal lasso portion of the catheter 72 is typically formed of shape-memory retentive material such as nitinol.

(22) It is understood that the balloon catheter 24 may also be supported by a linear or focal catheter 99 (as shown in broken lines in FIG. 2) in the PV or elsewhere in the heart. The focal catheter 99 may include a force sensor at its distal tip. Suitable force sending distal tips are disclosed in U.S. Pat. No. 8,357,152, issued on Jan. 22, 2013 to Govari et al., titled CATHETER WITH PRESSURE SENSING, and in U.S. Patent Application 2011/0130648, to Beeckler et al., filed Nov. 30, 2009, titled CATHETER WITH PRESSURE MEASURING TIP, the entire contents of both of which are incorporated herein by reference. Any catheter used in conjunction with the balloon catheter may have features and functions, including, for example, pressure sensing, ablation, diagnostic, e.g., navigation and pacing.

(23) An inflatable balloon 80 of the balloon catheter 24 has an exterior wall or membrane 26 of a bio-compatible material, for example, formed from a plastic such as polyethylene terephthalate (PET), polyurethane or PEBAX®. The shaft 70 and the distal shaft end 88 define a longitudinal

axis **78** of the balloon **80**. The balloon **80** is deployed, in a collapsed uninflated configuration, via the lumen **23** of the probe **20**, and may be inflated after exiting from the distal end **22**. The balloon **80** may be inflated and deflated by injection and expulsion of a fluid such as saline solution through the shaft **70**. The membrane **26** of the balloon **80** is formed with irrigation pores or apertures **27** (shown in FIG. 5) through which the fluid can exit from the interior of the balloon **80** to outside the balloon for cooling the tissue ablation site at the ostium. While FIG. 3 shows fluid exiting the balloon **80** as jet streams, it is understood that the fluid may exit the balloon with any desired flow rate and/or pressure, including a rate where the fluid is seeping out of the balloon.

(24) The membrane **26** supports and carries a combined electrode and temperature sensing member which is constructed as a multi-layer flexible circuit electrode assembly **84**. The “flex circuit electrode assembly” **84** may have many different geometric configurations. In the illustrated embodiment, the flex circuit electrode assembly **84** has a plurality of radiating leaves or strips **30**.

(25) FIG. 4 is a schematic view of a plurality of leaves **30**, according to an embodiment of the present invention. The leaves **30** are evenly distributed about the distal end **88** and the balloon **80**. Each leaf has wider proximal portion that gradually tapers to a narrower distal portion. FIG. 4 shows, by way of example, ten radiating leaves **30**, but it will be understood that embodiments of the present invention may have more or fewer than ten leaves. In one embodiment, referred to below, there are twelve leaves **30**.

(26) With reference to FIGS. 2 and 4, each leaf **30** has a proximal tail **31P** and a distal tail **31D**. The proximal tail **31P** is tucked under and fastened to the catheter **24** by a proximal ring **28P** mounted on the proximal shaft portion **82** of the shaft **70**. The distal tail **31D** is tucked under and fastened to the catheter **24** by a distal ring (not shown). Either or both sets of tails **31D** and **31P** may be further covered by a respective semispherical cap, such as distal cap **28D**. One or more contact electrodes **33** on each leaf come into galvanic contact with the ostium **11** during an ablation procedure, during which electrical current flows from the contact electrodes **33** to the ostium **11**, as shown in FIG. 3. In the description, electrodes **33** are differentiated as necessary by appending a letter to the identifying numeral, so that there are electrodes **33A**, **33B**,

(27) FIG. 5 is a schematic perspective view of a flexible circuit assembly partly lifted from the balloon catheter according to an embodiment of the present invention. For simplicity, the flex circuit electrode assembly **84** is described with respect to one of its leaves **30** as shown in FIG. 5, although it is understood that the following description may apply to each leaf of the assembly. The flex circuit electrode assembly **84** includes a flexible and resilient sheet substrate **34**, constructed of a suitable bio-compatible materials, for example, polyimide. In some embodiments, the sheet substrate **34** has a greater heat resistance (or a higher melting temperature) compared to that of the balloon membrane **26**. In some embodiments, the substrate **34** is constructed of a thermoset material having a decomposition temperature that is higher than the melting temperature of the balloon membrane **26** by approximately 100° C. or more.

(28) The substrate **34** is formed with one or more irrigation pores or apertures **35** that are in alignment with the irrigation apertures **35** of the balloon member **26** so that fluid passing through the irrigation apertures **35** can pass to the ablation site on the ostium.

(29) The substrate **34** has a first or outer surface **36** facing away from the balloon membrane **26**, and a second or inner surface **37** facing the balloon membrane **26**. On its outer surface **36**, the substrate **34** supports and carries the contact electrodes **33** adapted for tissue contact with the ostium. On its inner surface **37**, the substrate **34** supports and carries a wiring electrode **38**. The contact electrode **33** delivers RF energy, supplied by power supply **54**, to the ostium during ablation and is connected to a thermocouple junction (described in more detail below) for temperature sensing of the ostium. In the illustrated embodiment, the contact electrode **33** has a longitudinally elongated portion **40** and a plurality of thin transversal linear portions or fingers **41** extending generally perpendicularly from each lateral side of the elongated portion **40** between enlarged proximal and distal ends **42P** and **42D**, generally evenly spaced therebetween. The

elongated portion **40** has a greater width and each of the fingers has a generally uniform lesser width. Accordingly, the configuration or trace of the contact electrode **33** resembles a “fishbone.” (30) Formed within the contact electrode **33** are one or more exclusion zone **47**, each surrounding an irrigation aperture **27** formed in the substrate **26**. The exclusion zones **47** are voids purposely formed in the contact electrode **33**, as explained in detail further below, so as to avoid damage to the contact electrode **33** during construction of the electrode assembly **84** in accommodating the irrigation apertures **27** at their locations and in their function.

(31) Also formed in the contact electrode **33** are one or more conductive blind vias **48** which are conductive or metallic formations that extend through through-holes in the substrate **34** and are configured as electrical conduits connecting the contact electrode **33** on the outer surface **36** and the wiring electrode **38** on the inner surface **37**. It is understood that “conductive” is used herein interchangeably with “metallic” in all relevant instances.

(32) Attached, e.g., by a solder weld **63**, to an active solder pad portion **61A** of electrode **38** are a wire pair, e.g., a constantan wire **51** and a copper wire **53**. The copper wire **53** provides a lead wire to the wiring electrode **33**, and the copper wire **53** and the constantan wire **51** provide a thermocouple whose junction is at solder weld **63**, so that weld **63** is also referred to herein as thermocouple junction **63**. Junction **63** acts as temperature sensor, and is also referred to herein as sensor **63**, and the sensors are differentiated as necessary by appending a letter to the identifying numeral, so that there are sensors **63A**, **63B**, Thus, for each electrode **33A**, **33B**, . . . , there is a respective temperature sensor **63A**, **63B**,

(33) The wire pair **51/53** is passed through a through-hole **29** formed in the membrane **26**. It is understood that, in other embodiments in the absence of the through-hole **29**, the wire pair **51/53** may run between the membrane **26** and the substrate **34** and further proximally between the membrane **26** and the proximal tail **31P** until the wire pair **51/53** enters the tubular shaft **70** via another through-hole (not shown) formed in the tubular shaft sidewall closer to the proximal ring **28**.

(34) The flex circuit electrode assembly **84**, including the leaves **30** and the tails **31P** and **31D**, is affixed to the balloon membrane **26** such that the outer surface **36** of the substrate **34** is exposed and the inner surface **37** of the substrate **34** is affixed to the balloon membrane **26**, with the wiring electrode **38** and wire pair **51/53** sandwiched between the substrate **34** and the balloon membrane **26**.

First Embodiment

(35) FIG. **6** is a block diagram of apparatus **12**, according to a first embodiment of the present invention. In FIG. **6** processor **46**, temperature module **52**, power supply **54**, switch **57**, and catheter **24** are illustrated as rectangular blocks, and the block diagram also illustrates sensing signals, control signals, and power connections between the different elements of apparatus **12**. Switch **57** comprises a plurality of sub-switches **59A**, . . . , **59D**, . . . , **59N**, collectively termed sub-switches **59**. Catheter **24** comprises electrodes **33A**, . . . , **33D**, . . . **33N**, which are respectively attached to sensors **63A**, . . . , **63D**, . . . **63N**, and which are also connected to receive power from the power supply via sub-switches **59A**, . . . , **59D**, . . . , **59N**. In the first embodiment of apparatus **12** illustrated by FIG. **6**, all sub-switches **59** are constantly closed, so that, when activated, power supply **54** supplies power simultaneously to all electrodes **33**.

(36) Thus, in operation of apparatus **12**, and referring also to FIG. **1**, temperature module **52** receives sensing signals from each sensor **63** of each electrode **33**, and uses the signals to determine a tissue temperature, which is the temperature of the tissue surface in contact with each of the electrodes. The temperature module is configured to calculate the tissue temperature at a fixed rate, herein assumed to be every 33 ms, but other embodiments may calculate the tissue temperature at higher or lower rates. The temperature module passes the calculated tissue temperature values for each of electrodes **33** to processor **46**, which in turn passes control signals to power supply **54**.

(37) Power supply **54** provides RF power, separately and individually, via respective sub-switches

59, to each electrode 33 of balloon catheter 24. In some embodiments the RF power is provided via copper wire 53. Alternatively or additionally, the RF power may be provided to the respective electrodes 33 by another conductor. The power for each electrode may be supplied in a range of 1 W to 100 W, and the power may be provided simultaneously to all electrodes 33. Thus, in an embodiment of the invention comprising twelve electrodes 33, module 54 may supply 100 W to each electrode, for an overall power input to the catheter of 1.2 kW.

(38) In order to supply these high powers, it will be understood that leads to electrodes 33, and substrate 34, provide sufficient insulation so as to avoid any arcing from the electrodes.

(39) In embodiments of the present invention the power supply can be configured to provide a maximum RF power to each electrode 33 that can be set within a range of 70 W-100 W. In some embodiments, the module can be configured to provide a further RF power to each electrode 33 in a different range from the maximum. In one embodiment the further power range is 20 W-60 W, and the further power is typically provided after the maximum power. The maximum RF power and the further RF power are also termed herein the first power and the second power.

(40) The power supply also measures an impedance of each electrode 33. The impedance is measured at a predefined rate, herein assumed to be every 500 ms, but other embodiments may measure the impedance at a lower or higher rate.

(41) For each electrode 33, the maximum power for the electrode, and the time period for which the power is delivered, is selected by professional 14. The professional may also select values of the power less than 70 W, and corresponding time periods for delivery of this reduced power. The actual power delivered by any given electrode is determined by the tissue temperature received from temperature module 52 for that electrode, as described below.

(42) Typically, during an ablation session, the impedance presented to a given electrode 33 decreases. Embodiments of the present invention also check if the impedance presented to each electrode increases from a previous impedance measurement by more than a pre-set value, herein assumed to be 7Ω , although other embodiments may use larger or smaller values of impedance for the pre-set value. An increase of impedance typically occurs if there is an unwanted change in the tissue being ablated, such as charring or steam popping. If, for any given electrode 33, the impedance increases by more than the pre-set value, the power supply is configured to stop the RF delivery to the given electrode.

(43) Notwithstanding the powers selected by the professional, the power supply is configured to reduce the power delivered by a given electrode, typically by between approximately 5% and approximately 95%, if the tissue temperature for the given electrode, received from the temperature module, reaches or exceeds a predefined temperature threshold. The predefined temperature threshold is a maximum allowable temperature that is set by professional 14, and in the following description the predefined temperature threshold is also referred to as the maximum allowable temperature.

(44) In one embodiment, power for a given electrode that has been originally set to 90 W is reduced to 50 W after 4s, regardless of the reading from sensor 63. In an embodiment of the present invention, the maximum allowable temperature for all electrodes may be set within a range 60°C .- 65°C . Typically, exceeding the maximum allowable temperature causes undesirable effects such as charring, coagulation on an electrode 33, and/or steam pops in the tissue being ablated.

(45) Irrigation module 58 (FIG. 1) governs the rate at which irrigation fluid is delivered to balloon catheter 24. In embodiments of the present invention it may be set within the range of 5-60 ml/min.

(46) FIG. 7 is a flowchart of steps of an algorithm performed in operation of the first embodiment of apparatus 12. The steps of the flowchart assume that the block diagram of FIG. 6 applies, i.e., that all sub-switches 59 are constantly closed, so that power supply supplies power simultaneously to all electrodes 33.

(47) In a range setting step 200, ranges for each of the variable parameters referred to above are set. The ranges may be set individually for each electrode 33. While in one embodiment this is

typically the same for all electrodes, this is not necessarily the case, and in other embodiments ranges are different for different electrodes.

(48) In one embodiment the ranges are set as shown in Table I. Typically, for the powers, an operator of apparatus **12**, usually professional **14**, only sets the first power, while the second power is automatically pre-set by processor **46**.

(49) TABLE-US-00001 TABLE I Parameter Range Maximum Power Delivered 50 W-100 W (First Power) Second Power 15 W-50 W Maximum allowable 50° C.-65° C. temperature Irrigation rate 5-60 ml/min First Time Period (during 1 s to 6 s which First Target Power is operative) Second Time Period (during Up to 14 s which Second Target Power is operative) Overall Time Period for 1 s-20 s Power Delivery (Sum of First and Second Time Periods)

(50) Range setting step **200** is implemented before professional **14** performs an ablation.

(51) At the beginning of an ablation session, in a probe introduction step **202**, professional **14** inserts balloon catheter **24** into a desired location in myocardium **16**, using the tracking system incorporated into apparatus **12**.

(52) In a select value step **204**, prior to performing the ablation procedure, professional **14** selects values of the parameters listed in Table I that are to be used in the procedure, and uses controls **49** to provide the values to the system. Alternatively, the professional selects a predetermined set of the values of the parameters listed in Table I, typically by choosing a “recipe” comprising the values, from a group of such recipes. The selected values typically depend on the depth of lesion it is desired to form by the procedure. For lesions of 1-3 mm depth the inventors have found that the values of the parameters given by Table II give good results. For lesions of 4-5 mm depth the inventors have found that the values of the parameters given by Table III give good results.

(53) It will be understood that the selections made by professional **14** in step **204** are for each electrode **33** individually. Thus for twelve electrodes **33**, twelve sets of parameters are selected. While the selections may be the same for all electrodes **33**, this is not a requirement. For example, the professional may select parameters according to Table II for some electrodes, and according to Table III for other electrodes.

(54) In addition, while the selected parameters for each electrode **33** are typically applied simultaneously to all the electrodes, this is also not a requirement. Thus the parameters may be provided at least partially sequentially, and/or in a staggered fashion, e.g., 2s after ablation has been started for one electrode, ablation may be started for a neighboring electrode.

(55) TABLE-US-00002 TABLE II Lesions of 1-3 mm Depth Parameter Value First Target Power 90 W Second Target Power Not set Maximum allowable 60° C. temperature Irrigation rate 8 ml/min Time Period 4 s

(56) TABLE-US-00003 TABLE III Lesions of 4-5 mm Depth Parameter Value First Target Power 90 W Second Target Power 50 W Maximum allowable 60° C. temperature Irrigation rate 8 ml/min First Time Period 4 s Second Time Period 6 s

(57) Those having skill in the art will be able to determine, for other lesion depths, required values of the parameters within the ranges given by Table I, without undue experimentation.

(58) In a begin RF delivery step **206**, professional **14** operates apparatus **12**, with the parameter values selected in step **204**, in order to perform the ablations of electrodes **33**. Typically, during the ablations, screen **62** displays values of the parameters listed in Table I to the professional. Screen **62** may also be configured to display to the professional, by methods which are known in the art, the progress of the RF deliveries to the individual electrodes. The display of the progress may be graphical, such as a simulation of the dimensions of the respective lesions as they are produced by the ablations, and/or alphanumeric.

(59) During the RF delivery procedure processor **46** uses the temperature module and the power supply to perform a number of checks on the progress of the procedure, as shown in the flowchart by decision steps **208**, **210**, and **214**.

(60) Processor **46** operates steps **208-222** of the algorithm for each given electrode **33** individually

and separately, measuring the impedance for the given electrode and the tissue temperature provided by sensor **63** of the electrode. For clarity, in the description below the ablations of all electrodes are assumed to be implemented simultaneously, in which case the processor performs steps **206-222** simultaneously for all the different electrodes. Those having skill in the art will be able to adapt the description, mutatis mutandis, for cases of non-simultaneous ablation.

(61) In step **208**, processor **46** uses power supply **54** to check if the impedance of a given electrode **33** has increased by more than the pre-set impedance value. If it has, the system halts the procedure for the given electrode in a termination step **216**. If step **208** returns a negative value, control of the algorithm continues to decision step **210**.

(62) In step **210**, the processor uses temperature module **52** to check if the measured tissue temperature for the given electrode, as measured by sensor **63** of the electrode, exceeds or reaches the predefined temperature threshold, i.e., the maximum allowable temperature selected in step **204**. If decision step **210** returns a positive value, the power supply, in a reduction step **218**, reduces the power to the given electrode.

(63) The power reduction in step **218** is a function of a number of parameters: A difference in temperature between the maximum allowable temperature T (set in step **204**) and the measured temperature $T_{sub.t}$ at a time t , A change of measured temperatures between sequential temperature measurements, i.e., $T_{sub.t-1} - T_{sub.t}$, A target power P , where if the flowchart is functioning in the first time period, P is the first target power, and if the flowchart is functioning in the second time period, P is the second target power, and A power $P_{sub.t}$ measured at time t .

(64) In one embodiment the following equations applies for the power reduction:

(65)
$$P(T) = \frac{a(T_{t-1} - T_t)}{T} + \frac{b(T - T_t)}{T} \quad (1)$$
 where $\Delta P(T)$ is a fractional change in power as a function of temperature, and a and b are numerical constants. In a disclosed embodiment $a=10$ and $b=1$.

(66)
$$P(p) = \frac{(P - P_t)}{P} \quad (2)$$
 where $\Delta P(p)$ is a fractional change in power as a function of power.

$\Delta P = \min(\Delta P(T), \Delta P(p)) \quad (3)$ where $\min(\Delta P(T), \Delta P(p))$ is the minimum of $\Delta P(T)$ and $\Delta P(p)$, and ΔP is the fractional change in power applied in step **218**.

(67) Typically, power reduction step **218** is performed reiteratively with decision step **210**, until the measured temperature is below the predefined temperature threshold.

(68) If step **210** returns a negative value, control continues to decision step **214**.

(69) In decision step **214**, processor **46** checks if the time for the ablation by the given electrode, set in step **204**, has been reached. If it has, then the flowchart ends. If the time has not been reached, control passes to a continuing ablation step **222**, where the processor continues the ablation by the given electrode, and returns to decision steps **208**, **210**, and **214**. Decision steps **208**, **210**, and **214** have been presented sequentially in the flowchart for simplicity and clarity. Typically, however, the system uses the power supply to perform the steps in parallel.

Second Embodiment

(70) FIG. **8** is a block diagram of apparatus **12**, according to a second embodiment of the present invention. Apart from the differences described below, the block diagram for the second embodiment is generally similar to that of the first embodiment (FIG. **6**) and elements indicated by the same reference numerals in both block diagrams are common. In contrast to the first embodiment, in the second embodiment of apparatus **12** illustrated by FIG. **6**, all sub-switches **59** are not constantly closed. Rather, as described in more detail below, during operation of apparatus **12** at any given instant some sub-switches **59** are open and some are closed. FIG. **8** illustrates, as an example, sub-switches **59A** and **59N** being open, while sub-switch **59D** is closed.

(71) FIG. **9** is a flowchart of steps of an algorithm performed in operation of the second embodiment of apparatus **12**.

(72) In contrast to the first embodiment described above, wherein power supply **54** is able to simultaneously supply ablation power to all electrodes **33**, in the second embodiment described hereinbelow the power supply module is limited to being only able to supply ablation power

simultaneously to a subset, i.e., a reduced number, of electrodes **33**, due to a maximum power rating of the power supply.

(73) In an initial step **250**, professional **14** inserts balloon catheter **24** into a desired location in myocardium **16**, using the tracking system incorporated into apparatus **12**.

(74) In an assignment step **254**, the professional assigns ablation parameters individually for each electrode **33**, i.e., a power to be delivered by each electrode, and a time duration for the delivery. While in some cases these may be the same for each electrode **33**, e.g. 90 W for 4 s for each electrode **33**, there is no requirement that this is the case. For example, electrode **33A** may be assigned 80 W for 3 s, electrode **33B** may be assigned 60 W for 4 s, electrode **33C** may be assigned 70 W for 3 s, and so on.

(75) In some embodiments the assignment may be determined by the positioning of catheter **24** (as implemented in step **250**) with respect to tissue being ablated. E.g., if balloon catheter **24** has been positioned to contact ostium **11** of pulmonary vein **13** (FIG. 3), the power and the time may be set according to a measured or assumed thickness of the ostium section contacting respective electrodes **33**.

(76) In addition to assigning ablation parameters individually for each electrode, in step **254** a predefined temperature threshold, for the temperature measured by sensors **63**, is assigned. The predefined temperature threshold may be assigned by professional **14**, or alternatively may be preset for apparatus **12**. The temperature threshold is, as stated above with reference to the first embodiment, a maximum allowable temperature for a given electrode **33**. As is also stated above, if the temperature of tissue becomes greater than the temperature threshold, the tissue may suffer undesired effects.

(77) In an initial ablation step **258**, the professional activates apparatus **12** to begin ablation. On activation, processor **46** divides electrodes **33** into two groups: a first group, herein termed active electrodes **33**, which are to be used for ablation, and a second group, herein termed quiescent electrodes **33**, which are not to be used for ablation. Processor **46** may make the selection by closing or opening sub-switches **59**. Thus, active electrodes **33** are selected by closing a first set of sub-switches **59**, so that there is a conducting line from power supply **54** to the electrodes for these electrodes. Quiescent electrodes **33** are selected by opening a second set of sub-switches **59**, so that there is no conducting line from power supply **54** to the electrodes for these electrodes. The processor stores identities of active electrodes **33** in an active electrode register, and identities of quiescent electrodes **33** in a quiescent electrode register.

(78) Processor **46** divides the electrodes into the two groups so that the total power required to be delivered to the active group, as determined according to the ablation parameters assigned in step **254**, does not exceed a maximum power rating of power module **54**. For example, if there are twelve electrodes **33**, each assigned in step **254** to deliver 50 W, and module **54** has a maximum power rating of 500 W, then the processor may assign up to five electrodes to be in the active group, and the remainder, in this case seven or more, to be in the quiescent group. Typically, processor **46** assigns the active group to have the largest possible number of electrodes **33**, consistent with the constraint that the maximum power rating of power module **54** is not exceeded. In some embodiments, the assignment may initially be made on a random basis. Alternatively, professional **14** may provide an indication to the processor of how the assignment is to be made on a non-random basis, for example, by providing the processor with a priority order of electrodes **33** to be assigned to the active group.

(79) In a continuation step **262**, processor **46** begins ablation by activating power supply **54**, so that the module supplies power to each of electrodes **33** of the active group, i.e., to each of the electrodes having respective sub-switches **59** closed, according to the power levels set in step **254**.

(80) In a first decision step **264**, processor **46** checks if a given active electrode **33** has completed its assigned ablation. For example, if in assignment step **254** a given electrode is assigned to ablate with a power 50 W for 4 s, the processor checks if an aggregate time during which the given

electrode is dissipating power of 50 W is equal to 4 s. If the first decision returns negative, then the processor proceeds to a temperature decision step **268**, where the processor checks, using the appropriate sensor **63**, if the given active electrode temperature exceeds the temperature threshold set in step **254**. If the threshold is not exceeded, i.e., the temperature condition returns negative, control returns to first decision step **264**.

(81) Thus, providing both first decision step **264** and temperature decision step **268** return negative, the processor iteratively loops through these decisions, and continues checking whether a given active electrode has completed its assigned ablation, and whether its temperature, as measured by its sensor, exceeds the temperature threshold.

(82) Processor **46** simultaneously implements the iterative loop described above for all active electrodes. The iterative loop follows a return line **270** of the flowchart.

(83) For any given active electrode, the iteration stops if either first decision step **264** or temperature decision step **268** returns positive, as is explained in more detail below.

(84) If temperature decision step **268** returns positive, i.e., if a sensor **63** of an active electrode **33** being checked indicates that the threshold temperature has been exceeded, control transfers to a switch toggling step **272**. Thus, if decision **268** returns positive, the iterative loop for the active electrode being checked is broken.

(85) In switch toggling step **272** processor **46** performs the following actions: Power to the active electrode checked prior to entry to step **272** is terminated, by the processor toggling in switch **57** the sub-switch **59** supplying the active electrode from a closed state to an open state. Using electrode identities in the quiescent electrode register, the processor selects a quiescent electrode to be converted to an active electrode. The selection is consistent with the constraint that the maximum power rating of power module **54**, when the new active electrode is operative, is not exceeded. The selected quiescent electrode is powered on, so becoming an active electrode, by the processor toggling in switch **57** the sub-switch **59** supplying the selected electrode from an open state to a closed state. The processor updates the active electrode register and quiescent electrode register accordingly. I.e., the identity of the electrode checked in decision step **268** is transferred from the active electrode register to the quiescent electrode register, and the identity of the quiescent electrode selected in the switch toggling step is transferred from the quiescent electrode register to the active electrode register.

(86) Once the processor has performed the actions described above for step **272**, control returns to decision step **264**, so that the iterative loop of decision steps **264**, **268**, and line **270** restarts.

(87) As stated above, the iterative loop may also terminate if decision step **264** returns positive. In this case the electrode being checked has completed its ablation, and control continues to a record completed electrodes step **276**. In step **276** the processor transfers the identity of the electrode checked in decision step **264** from the active electrode register to a completed ablation register.

(88) The flowchart continues to a decision step **280**, wherein the processor checks the completed ablation register to see if all electrodes have completed their ablation. If step **280** returns negative, i.e., there is at least one electrode that has not completed its assigned ablation, control continues to switch toggling step **272**, described above, wherein the active electrode checked in condition **264** is switched off, and an available quiescent electrode is switched on.

(89) If step **280** returns positive, i.e., all electrodes have completed their ablation, then the active electrode checked in decision step **264**, which is the last operative electrode, is switched off in a final flowchart step **284**, and the flowchart ends.

(90) While the description above has used a balloon catheter to provide multiple electrodes **33** for respective ablations, it will be understood that embodiments of the present invention are not limited to balloon catheters. Thus embodiments of the present invention comprise other catheters, such as basket catheters, lasso catheters, and focal catheters, with multiple electrodes that are used for respective ablations.

(91) It will thus be appreciated that the embodiments described above are cited by way of example,

and that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof which would occur to persons skilled in the art upon reading the foregoing description and which are not disclosed in the prior art.

Claims

1. An apparatus comprising: one or more processors; and a non-transitory computer readable medium storing a plurality of instructions, which when executed, cause the one or more processors to: cause a first set of switches respectively associated with a first set of electrodes on a catheter to be closed to establish a conducting line from a power supply to the first set of electrodes and a second set of switches respectively associated with a second set of electrodes on the catheter to be open so there is no conducting line from the power supply to the second set of electrodes, the power supply being configured to provide radiofrequency (RF) ablation energy via the catheter to an organ of a human body; cause the supply of RF ablation energy from the power supply to the first set of electrodes; cause each switch of the first set of switches to be opened to halt the supply of RF ablation energy when the associated electrode of the first set of electrodes completes its respective assigned ablation or a temperature of the associated electrode exceeds its respective predetermined temperature threshold as measured by a temperature sensor associated with that electrode; cause, in response to each switch of the first set of switches being opened, a switch of the second set of switches to be closed to establish a conducting line from the power supply to the electrode of the second set of electrodes associated with the closed switch; and cause each switch of the second set of switches to be opened to halt the supply of RF ablation energy when the associated electrode of the second set of electrodes completes its respective assigned ablation or a temperature of the associated electrode exceeds its respective predetermined temperature threshold as measured by a temperature sensor associated with that electrode.
2. The apparatus according to claim 1, wherein the one or more processors is configured to cause the first set of switches respectively associated with a first set of electrodes on a catheter to be closed to establish a conducting line from a power supply to the first set of electrodes so that the RF ablation energy required to be delivered to the first subset of electrodes does not exceed a maximum power rating of the power supply.
3. The apparatus according to claim 2, wherein the one or more processors is configured to cause, in response to each switch of the first set of switches being opened, a switch of the second set of switches to be closed to establish a conducting line from the power supply to the electrode of the second set of electrodes associated with the closed switch consistent with the determination that the RF ablation energy required to be delivered to the electrodes from the first and second subsets of electrodes does not exceed a maximum power rating of the power supply.
4. The apparatus according to claim 1, wherein the one or more processors is configured to store the identities of each electrode of the plurality of electrodes having a closed switch in an active electrode register and store the identities of each electrode of the plurality of electrodes having an open switch in a quiescent electrode register.
5. The apparatus according to claim 1, wherein the one or more processors is configured to update the active electrode register and the quiescent electrode register in response to each switch of the first set of switches being opened and correspondingly each switch of the second set of switches being closed.
6. The apparatus according to claim 1, wherein the catheter comprises a balloon catheter.
7. The apparatus according to claim 1, wherein the catheter comprises a basket catheter.
8. A method, comprising: causing a first set of switches respectively associated with a first set of electrodes on a catheter to be closed to establish a conducting line from a power supply to the first

set of electrodes and a second set of switches respectively associated with a second set of electrodes on the catheter to be open so there is no conducting line from the power supply to the second set of electrodes, the power supply being configured to provide radiofrequency (RF) ablation energy via the catheter to an organ of a human body; causing the supply of RF ablation energy from the power supply to the first set of electrodes; causing each switch of the first set of switches to be opened to halt the supply of RF ablation energy when the associated electrode of the first set of electrodes completes its respective assigned ablation or a temperature of the associated electrode exceeds its respective predetermined temperature threshold as measured by a temperature sensor associated with that electrode; causing, in response to each switch of the first set of switches being opened, a switch of the second set of switches to be closed to establish a conducting line from the power supply to the electrode of the second set of electrodes associated with the closed switch; and causing each switch of the second set of switches to be opened to halt the supply of RF ablation energy when the associated electrode of the second set of electrodes completes its respective assigned ablation or a temperature of the associated electrode exceeds its respective predetermined temperature threshold as measured by a temperature sensor associated with that electrode.

9. The method according to claim 8, wherein causing the first set of switches respectively associated with a first set of electrodes on a catheter to be closed to establish a conducting line from a power supply to the first set of electrodes is based upon a determination that the RF ablation energy required to be delivered to the first subset of electrodes does not exceed a maximum power rating of the power supply.

10. The method according to claim 9, wherein causing, in response to each switch of the first set of switches being opened, a switch of the second set of switches to be closed to establish a conducting line from the power supply to the electrode of the second set of electrodes associated with the closed switch is consistent with the determination that the RF ablation energy required to be delivered to the electrodes from the first and second subsets of electrodes does not exceed a maximum power rating of the power supply.

11. The method according to claim 8, further comprising storing the identities of each electrode of the plurality of electrodes having a closed switch in an active electrode register and storing the identities of each electrode of the plurality of electrodes having an open switch in a quiescent electrode register.

12. The method according to claim 11, further comprising updating the active electrode register and the quiescent electrode register in response to each switch of the first set of switches being opened and correspondingly each switch of the second set of switches being closed.

13. The method according to claim 8, wherein the catheter comprises a balloon catheter.

14. The method according to claim 8, wherein the catheter comprises a basket catheter.

15. A computer program product, comprising a non-transitory computer-readable medium having computer-readable program code embodied therein to be executed by one or more processors, the program code including instructions to: cause a first set of switches respectively associated with a first set of electrodes on a catheter to be closed to establish a conducting line from a power supply to the first set of electrodes and a second set of switches respectively associated with a second set of electrodes on the catheter to be open so there is no conducting line from the power supply to the second set of electrodes, the power supply being configured to provide radiofrequency (RF) ablation energy via the catheter to an organ of a human body; cause the supply of RF ablation energy from the power supply to the first set of electrodes; cause each switch of the first set of switches to be opened to halt the supply of RF ablation energy when the associated electrode of the first set of electrodes completes its respective assigned ablation or a temperature of the associated electrode exceeds its respective predetermined temperature threshold as measured by a temperature sensor associated with that electrode; cause, in response to each switch of the first set of switches being opened, a switch of the second set of switches to be closed to establish a conducting line

from the power supply to the electrode of the second set of electrodes associated with the closed switch; and cause each switch of the second set of switches to be opened to halt the supply of RF ablation energy when the associated electrode of the second set of electrodes completes its respective assigned ablation or a temperature of the associated electrode exceeds its respective predetermined temperature threshold as measured by a temperature sensor associated with that electrode.

16. The computer program product according to claim 15, wherein the program code includes further instructions to cause the first set of switches respectively associated with a first set of electrodes on a catheter to be closed to establish a conducting line from a power supply to the first set of electrodes is based upon a determination that the RF ablation energy required to be delivered to the first subset of electrodes does not exceed a maximum power rating of the power supply.

17. The computer program product according to claim 16, wherein the program code includes further instructions to cause, in response to each switch of the first set of switches being opened, a switch of the second set of switches to be closed to establish a conducting line from the power supply to the electrode of the second set of electrodes associated with the closed switch consistent with the determination that the RF ablation energy required to be delivered to the electrodes from the first and second subsets of electrodes does not exceed a maximum power rating of the power supply.

18. The computer program product according to claim 15, wherein the program code includes further instructions to store the identities of each electrode of the plurality of electrodes having a closed switch in an active electrode register and store the identities of each electrode of the plurality of electrodes having an open switch in a quiescent register.

19. The computer program product according to claim 15, wherein the program code includes further instructions to update the active electrode register and the quiescent electrode register in response to each switch of the first set of switches being opened and correspondingly each switch of the second set of switches being closed.

20. The computer program according to claim 15, wherein the catheter comprises one of a balloon catheter or a basket catheter.
